

YING SHU

3611 Washington St, Boston, MA, 02130 | (857)-294-9127 | yingshu0924@gmail.com

EDUCATION

University of Pennsylvania

Candidate for Master of Science in Data Analytics & Social Policy

Philadelphia, PA
Aug 2022 – Dec 2023

Awards: *Second place* in 2023 Georgetown Green Space Data Challenge; **GPA:** 3.98/4.0

Relevant Courses: Applied Machine Learning, Big Data Analytics, Applied Linear Regression, Research and Evaluation

Northeastern University

Bachelor of Science in Marketing Analytics & Finance

Boston, MA
Aug 2019 - May 2022

Awards: Dean's List (2020 Spring, 2020 Fall, 2021 Spring, 2021 Fall, 2022 Spring); **GPA:** 3.63/4.0

Relevant Courses: Financial Analytics with Python, Business Statistics, Marketing Analytics, Marketing Research

SKILLS

- **Programming Languages:** SQL, Python, R; **Tools:** Tableau, Power BI, Smartsheet, Salesforce, Excel, Google Analytics, SPSS, ArcGIS, Jira, Metabase, Asana, Confluence;
- **Technical:** Data Management, NLP, A/B testing, Statistics, Machine Learning; **Business:** Product Management, Marketing, PMO, Survey Design, Project Management, Agile Methodology (Scrum, Kanban), Documentation; **Language:** Mandarin, English

RELEVANT PROFESSIONAL EXPERIENCE

Mercy Corps

Data Analyst Intern

Portland, OR
May 2023 - Sep 2023

- **Led, designed and analyzed** business outcome **survey** with 2,000+ records using **R**, performing hypothesis testing to understand user feedbacks and presented the findings to the **executive team**
- Leveraged **Naïve Bayes** in **Python** to perform **text mining**, processing 10,000+ messages on the platform to extract meaningful user behavior patterns, and conducted **sentiment analysis** using **Python (NLTK)**, visualizing results with **word clouds**
- Implemented **K-means clustering** to identify and profile top 10 active user clusters (**customer segmentation**) to understand the user population
- Facilitated data-driven decision-making by executing 50+ data requests using **SQL, Python** and **R**, maintaining high-quality **dashboards** and reports
- Guaranteed replicability, standardization, and **data quality** by **documenting data analysis processes** and maintaining the **Metabase database**

Harvard Business Publishing

PMO Analyst Intern

Boston, MA
Jul 2021 - Dec 2021

- Saved 3 hours per week of manual reporting by creating a template in **Excel** used to bulk upload client info on **Salesforce**
- Pinpointed internal inefficiencies and updated program plans by developing 5 **PM summary reports** using **Excel pivot tables**
- Guaranteed **standardized business process** by consolidating project reporting materials in **Confluence** and **Smartsheet** for monthly review by PMs
- Conducted monthly audits of data for 1000+ active and inactive users across different platforms using **SQL** queries, ensuring information consistency across multiple enterprise platforms

RELEVANT RESEARCH&PROJECT EXPERIENCE

Credit Score Classification (Machine Learning)

Philadelphia, PA

- Conducted EDA and data preprocessing, created data visualizations, and employed **feature selection** methods in **Python**
- Implemented **supervised learning** and **deep learning** algorithms like **Feedforward Neural Network (FNN)** to classify credit scores and predict creditworthiness
- Applied **cross validation** for rigorous evaluation and utilized **hyperparameter tuning** to optimize model performance, the final model achieving a classification rate of 90%
- Evaluated the models using different techniques including **ROC curves**

Greenspace Data Challenge for Health Implications (2nd place win, Data Analysis/GIS)

Philadelphia, PA

- Leveraged **ArcGIS** and **SQL queries** to create insightful **geo-social visualizations** of greenness in Philadelphia
- Utilized **R** programming to build linear models of the health implications associated with greenspace coverage
- Formulated **10-page report** and presented impactful recommendation including allocating funding to vulnerable neighborhoods

Live Like a Local Tour Boston (Market Research)

Boston, MA

- Designed a questionnaire using **Qualtrics** to gain insights into the Boston tourism market's demographics and preferences.
- Analyzed the survey results using **SPSS (Chi-square, ANOVA)** and **Python (Pandas)**
- Provided the client with strategic recommendations to **enhance ad engagement** with college students and promote family packages.

Factors Affecting Quitting Electronic Nicotine Delivery System (ENDS) (Statistics)

Philadelphia, PA

- Built a **linear regression model** with **interaction effect** in **R** between perceived difficulty of quitting ENDS and number of attempts to quit and used different metrics to evaluate the model
- Built a **logistic regression model** in **R** examining the use of ENDS within the past 30 days of data collection and the number of attempts to quit using ENDS, which had meaningful policy implications
- Formulated a **40-page report** based on the findings